

AVERAGING OF THE PROBLEM OF OPTIMAL CONTROL ON TIME SCALES

A. P. Ogulenko and O. D. Kichmarenko

UDC 517.9

A numerical-asymptotic method for the solution of the problem of optimal control over the system on time scales is substantiated.

Introduction

The outstanding results obtained by Krylov and Bogolyubov [1, 2] not only laid the foundations of new theoretical investigations but also have been extensively used in various branches of applied science. Moiseev [3] was the first who applied the method of averaging to the analysis of the problems of optimal control. In the works by Plotnikov and his colleagues, this method was applied to new classes of the problems of control over systems with delay, with discontinuous right-hand sides and pulses, with multivalued and fuzzy right-hand sides, systems with discrete time, etc. [4–13]. The indicated variety of the mathematical models is explained by the necessity of investigation of complex processes with noticeably different specific features.

The approach to understanding the nature of time in dynamical systems underwent principal changes after the appearance of Hilger's work [14]. The notion of time scale allowed one to construct the general theory of dynamical systems capable of description not only of continuous and discrete systems but also, which is much more important, of the mixed cases. The detailed presentation of the theory of dynamical systems on time scales can be found in [15, 16].

The method of averaging for dynamical systems on time scales was presented in the work [17] devoted to the investigation of the problem of the closeness of the solution of the original system on time scale to the solution of the averaged generalized differential equation.

In [18], we substantiated the scheme of complete averaging (an analog of the Bogolyubov theorem) under the conditions that the averaged system has the same nature as the original system and both dynamical systems are defined on the same time scale.

The aim of the present work is to construct the method of averaging for controlled systems on time scales, to determine the algorithm of correspondence of controls over the original and averaged systems, and finally, to substantiate the numerical-asymptotic method for the solution of the problem of control over the system on time scale.

Auxiliary Information

We now present some basic information about time scales required for the presentation of the accumulated results. We introduce and denote the definitions and concepts according to [15, 16].

A time scale is defined as a nonempty closed subset of the set of real numbers and denoted by $\mathbb{T}$. The properties of the time scale are determined by the following three functions:

(i) the forward-shift operator:

$$\sigma(t) = \inf \{s \in \mathbb{T} : s > t\};$$

Mechnikov Odessa National University, Dvoryanskaya Str., 2, Odessa, 65026, Ukraine.

Translated from *Nelineini Kolyvannya*, Vol. 17, No. 3, pp. 365–378, July–September, 2014. Original article submitted September 6, 2013; revision submitted January 22, 2014.

(ii) the backward-shift operator:

$$\rho(t) = \sup \{s \in \mathbb{T} : s < t\}$$

(in this case, we set $\inf \emptyset = \sup \mathbb{T}$ and $\sup \emptyset = \inf \mathbb{T}$);

(iii) the granularity function

$$\mu(t) = \sigma(t) - t.$$

The behavior of the forward- and backward-shift operators at a given point of the time scale specifies the type of this point. The corresponding classification of points of the time scale is presented in Table 1.

t dispersed on the right	$t < \sigma(t)$
t dense on the right	$t = \sigma(t)$
t dispersed on the left	$\rho(t) < t$
t dense on the left	$\rho(t) = t$
t isolated	$\rho(t) < t < \sigma(t)$
t dense	$\rho(t) = t = \sigma(t)$

We define a set $\mathbb{T}^\kappa$ in the following way:

$$\mathbb{T}^\kappa = \begin{cases} \mathbb{T} \setminus \{M\}, & \text{if there exists a point dispersed on the right } M \in \mathbb{T} : \\ & M = \sup \mathbb{T}, \sup \mathbb{T} < \infty, \\ \mathbb{T}, & \text{otherwise.} \end{cases}$$

In what follows, we set

$$[a, b] = \{t \in \mathbb{T} : a \leq t \leq b\}.$$

Definition 1. The δ -partition of an interval $[t_0, t_f]$ on the time scale is defined as follows: For a given diameter of the partition δ , the first point of partition is set equal to t_0 and the subsequent points are given by the formula

$$t_i = \begin{cases} \sup(t_{i-1}, t_{i-1} + \delta), & \text{for } t_{i-1} + \delta \in \mathbb{T}^\kappa, \\ \sigma(t_{i-1}), & \text{for } t_{i-1} + \delta \notin \mathbb{T}^\kappa. \end{cases}$$

Remark 1. It is possible to show [16, p. 120] that the partition constructed in this way has the following properties: For any i , $t_i - t_{i-1} \leq \delta$ (we assume that, in this case, $i \in I_\delta$) or $t_i - t_{i-1} > \delta$ and $\sigma(t_{i-1}) = t_i$ (in this case, $i \in I_\sigma$). In what follows, we set $N = |I_\delta| + |I_\sigma|$. In other words, N is the number of points of the δ -partition. In addition, we denote by $I = I_\delta \cup I_\sigma$ the set of indices of all points of the δ -partition.

Definition 2 [15]. Let $f : \mathbb{T} \rightarrow \mathbb{R}$ and let $t \in \mathbb{T}^\kappa$. The number $f^\Delta(t)$ is called the Δ -derivative of a function f at the point t if, for any $\varepsilon > 0$, there exists a neighborhood U of the point t (i.e., $U = (t - \delta, t + \delta) \cap \mathbb{T}, \delta < 0$) such that

$$|f(\sigma(t)) - f(s) - f^\Delta(t)(\sigma(t) - s)| \leq \varepsilon |\sigma(t) - s| \quad \forall s \in U.$$

Definition 3 [15]. If $f^\Delta(t)$ exists for any $t \in \mathbb{T}^\kappa$, then $f : \mathbb{T} \rightarrow \mathbb{R}$ is called Δ -differentiable on $\mathbb{T}^\kappa$. The function $f^\Delta(t) : \mathbb{T}^\kappa \rightarrow \mathbb{R}$ is called the delta-derivative of a function f on $\mathbb{T}^\kappa$.

If f is differentiable with respect to t , then

$$f(\sigma(t)) = f(t) + \mu(t)f^\Delta(t).$$

Definition 4 [15]. The function $f : \mathbb{T} \rightarrow \mathbb{R}$ is called regular if it has finite right limits at all points of the time scale $\mathbb{T}$ that are dense on the right and finite left limits at all points dense on the left.

Definition 5 [15]. The function $f : \mathbb{T} \rightarrow \mathbb{R}$ is called rd -continuous if it is continuous at the points that are dense on the right and has finite left limits at the points that are dense on the left. The set of these functions is denoted by $C_{rd} = C_{rd}(\mathbb{T})$ and the set of differentiable functions whose derivative is rd -continuous is denoted by $C_{rd}^1 = C_{rd}^1(\mathbb{T})$.

Definition 6 [15]. For any regular function $f(t)$, there exists a function F differentiable in the domain D and such that the following equality holds for all $t \in D$:

$$F^\Delta(t) = f(t).$$

This function is defined ambiguously. It is called the preprimitive of $f(t)$.

The indefinite integral on the time scale takes the form

$$\int f(t)\Delta t = F(t) + C,$$

where C is any integration constant and $F(t)$ is the preprimitive for $f(t)$. If the relation $F^\Delta(t) = f(t)$, where $f : \mathbb{T} \rightarrow \mathbb{R}$ is an rd -continuous function, is true for all $t \in \mathbb{T}^\kappa$, then $F(t)$ is called the primitive of the function $f(t)$. If $t_0 \in \mathbb{T}$, then

$$F(t) = \int_{t_0}^t f(s)\Delta s$$

for all t . For all $r, s \in \mathbb{T}$, the definite Δ -integral is defined as follows:

$$\int_r^s f(t)\Delta t = F(s) - F(r).$$

Definition 7 [15]. Let $\mathbb{T}$ be a time scale and let X be a Banach space. A function $f : \mathbb{T} \times X \rightarrow X$ is called

- (i) *rd*-continuous if the function $g(t) = f(t, x(t))$ is *rd*-continuous for any continuous function $x : \mathbb{T} \rightarrow X$;
- (ii) *bounded in the domain* $Q \subset \mathbb{T} \times X$, if there exists a constant $M > 0$ such that $\|f(t, x)\| \leq M$ for any point $(t, x) \in Q$;
- (iii) *Lipschitz with respect to* x *in the domain* $Q \subset \mathbb{T} \times X$ if there exists a constant $\lambda > 0$ such that

$$\|f(t, x_1) - f(t, x_2)\| \leq \lambda \|x_1 - x_2\| \quad \forall (t, x_1), (t, x_2) \in Q.$$

Definition 8 [15]. A function $p : \mathbb{T} \rightarrow \mathbb{R}$ is called *regressive* if

$$1 + \mu(t)p(t) \neq 0, \quad t \in \mathbb{T}^\kappa.$$

The set of regressive and *rd*-continuous functions is denoted by $\mathcal{R} = \mathcal{R}(\mathbb{T})$.

As shown in [15], a function p from the class $\mathcal{R}$ can be associated with a function $e_p(t, s)$, which is an analog, by its properties, of the exponential function defined on $\mathbb{R}$.

In what follows, we also need the Gronwall inequality. It is formulated in the form of the following assertion:

Lemma 1 (Gronwall inequality [15]). Let y be a function *rd*-continuous on $\mathbb{T}$ and let $p \in \mathcal{R}^+$, $p \geq 0$, and $\alpha \in \mathbb{R}$. If the inequality

$$y(t) \leq \alpha + \int_{t_0}^t y(\tau)p(\tau)\Delta\tau, \quad \tau \in \mathbb{T}$$

holds, then

$$y(t) \leq \alpha e_p(t, t_0), \quad \tau \in \mathbb{T}.$$

Main Results

Consider the problem of optimal control over the system on the time scale

$$\begin{aligned} x^\Delta(t) &= \varepsilon[f(t, x(t)) + A(x(t))\varphi(t, u(t))], \\ x(t_0) &= x_0, \end{aligned} \tag{1}$$

with the terminal quality functional

$$\min J[u] = \Phi(x(t_1)). \tag{2}$$

Here, $\mathbb{T}$ is the time scale, i.e., a nonempty closed subset from $\mathbb{R}$, $\sup \mathbb{T} = +\infty$, $t \in \mathbb{T}$ is time, $x \in \mathbb{R}^n$, x^Δ is the Δ -derivative, $\varepsilon > 0$ is a small parameter, $f(t, x)$ is an n -dimensional vector function, $A(x)$ is an $(n \times m)$ -matrix, $\varphi(t, u)$ is an m -dimensional vector function, $\Phi(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$, $u(t) \in U$ is the r -dimensional control vector, $U \in \text{comp}(R^r)$, and $t_0 < \inf \mathbb{T}$.

We associate problem (1), (2) with the averaged problem of optimal control over the system on the same time scale

$$\xi^\Delta = \varepsilon[\bar{f}(\xi) + A(\xi)v], \quad (3)$$

$$\xi(t_0) = x_0$$

with the quality criterion

$$\min \bar{J}[v] = \Phi(\xi(t_1)). \quad (4)$$

The procedure of averaging is performed in the following way:

$$\bar{f}(x) = \lim_{\substack{T \rightarrow \infty \\ T \in \mathbb{T}}} \frac{1}{T} \int_{t_0}^T f(t, x) \Delta t, \quad (5)$$

$$v \in V, \quad V = \text{co} \left(\lim_{\substack{T \rightarrow \infty \\ T \in \mathbb{T}}} \frac{1}{T} \int_{t_0}^T \varphi(t, U) \Delta t \right).$$

The last integral in (5) is understood as an analog of the Aumann integral on the time scale and the convergence of the corresponding limit is understood in a sense of the Hausdorff metric.

We now consider an algorithm that enables one to establish the correspondence between the admissible controls in problems (1), (2) and (3), (4). First, we assume that the admissible control $v(t) \in V$ is given in the averaged problem. We associate it with the control $u(t) \in U$ in the original problem in the following way:

- (i) we select a δ -partition of the time scale $\mathbb{T}$ whose points are denoted by t_i ;
- (ii) find the points

$$v_i = \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} v(s) \Delta s, \quad i = 0, 1, 2, \dots;$$

- (iii) construct a control

$$u(t) = \{u_i(t), t_i \leq t < t_{i+1}\},$$

where $u_i(t)$ is a solution of the problem of minimization of the norm

$$\left\| \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \varphi(t, u(t)) \Delta t - v_i \right\|,$$

i.e.,

$$u_i(t) = \arg \min_{u(t) \in U} \left\| \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \varphi(t, u(t)) \Delta t - v_i \right\|.$$

Remark 2. It is possible to show that the set of points

$$\left\{ \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \varphi(t, u(t)) \Delta t, u(t) \in U \right\}$$

is a convex compact set.

Remark 3. The control $u_i(t)$ is determined, generally speaking, ambiguously.

Conversely, the control $u(t) \in U$ of the input problem is associated with the control $v(t) \in V$ of the averaged problem in the following way:

- (i) we fix the δ -partition of $\mathbb{T}$ defined above;
- (ii) compute the points

$$w_i = \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \varphi(t, u_i(t)) \Delta t, \quad i = 0, 1, 2, \dots;$$

- (iii) construct the control $v(t) = \{v_i, t_i \leq t < t_{i+1}\}$, where $v_i(t)$ is determined from the condition

$$v_i = \arg \min_{v \in V} \|w_i - v\|.$$

Remark 4. The control v_i is determined, generally speaking, ambiguously.

The following assertion substantiates the applicability of the method of averaging to the equations of controlled motion of systems (1) and (3).

Theorem 1. Let the following conditions be satisfied in the domain $Q = \{t \in \mathbb{T}, x \in D \subset \mathbb{R}^n, u \in U \subset \mathbb{R}^r\}$:

- (i) the function $f(t, x)$ is rd-continuous in t ; it is regressive and satisfies the conditions of existence and uniqueness of solution of the Cauchy problem; moreover, $\|f(t, x)\| \leq M$, $M > 0$, for any $(t, x) \in Q$ and $f(t, x)$ is Lipschitz with respect to x with constant $\lambda > 0$;
- (ii) the function $A(x)$ satisfies the Lipschitz condition in x with constant λ_A and is bounded by a constant M ;
- (iii) the function $\varphi(t, u)$ is rd-continuous in t , continuous in u , and bounded by a constant M ;
- (iv) for any admissible controls $v(t)$, the solution $\xi(t)$ of the averaged system (3) with initial condition $\xi(t_0) = x_0 \in D' \subset D$ belongs to the domain D together with its ρ -neighborhood;

(v) there exists a number $\mu_0 > 0$ such that $\mu(t) = 0$ or $\mu(t) > \mu_0$ for any $t \in \mathbb{T}^\kappa$.

Then, for any $0 < \eta \leq \rho$ and $L > 0$, there exist $\varepsilon_0(\eta, L) > 0$ and $\delta_0 > 0$ such that the following assertions are true for $0 < \varepsilon < \varepsilon_0$ and $t_0 \leq t \leq L\varepsilon^{-1}$:

(i) For any admissible control $v(t)$ of system (3), there exists a control $u(t)$ of system (1) such that the following inequality holds:

$$\|x(t) - \xi(t)\| \leq \eta, \quad (6)$$

where $x(t)$ is a solution of system (1) generated by the control $u(t)$ and $\xi(t)$ is a solution of system (3) generated by the control $v(t)$.

(ii) For any admissible control $u(t) \in U$ of system (1), there exists a control $v(t)$ of system (3) such that inequality (6) holds.

Proof. We now prove the first assertion of the theorem. Let $v(t)$ be an admissible control over system (3). According to the algorithm presented above, we associate this control with a control $u(t)$. The trajectory of motion of system (1) corresponding to the control $u(t)$ is denoted by $x(t)$, whereas the trajectory of motion of the averaged system under the action of the control $v(t)$ is denoted by $\xi(t)$.

We now estimate the closeness of the trajectories $x(t)$ and $\xi(t)$ in an interval of the time scale with length of order ε^{-1} :

$$\begin{aligned} \|x(t) - \xi(t)\| &= \left\| x_0 + \varepsilon \int_{t_0}^t [f(s, x(s)) - A(x(s))\varphi(s, u(s))] \Delta s \right. \\ &\quad \left. - x_0 - \varepsilon \int_{t_0}^t [\bar{f}(\xi(s)) - A(\xi(s))v(s)] \Delta s \right\| \\ &\leq \varepsilon \left\| \int_{t_0}^t [f(s, x(s)) - f(s, \xi(s))] \Delta s + \int_{t_0}^t [f(s, \xi(s)) - \bar{f}(\xi(s))] \Delta s \right. \\ &\quad \left. + \int_{t_0}^t [A(x(s)) - A(\xi(s))] \varphi(s, u(s)) \Delta s + \int_{t_0}^t A(\xi(s)) [\varphi(s, u(s)) - v(s)] \Delta s \right\| \\ &\leq \underbrace{\varepsilon \int_{t_0}^t \|f(s, x(s)) - f(s, \xi(s))\| \Delta s}_{B_1} + \underbrace{\varepsilon \int_{t_0}^t \|f(s, \xi(s)) - \bar{f}(\xi(s))\| \Delta s}_{B_2} \\ &\quad + \underbrace{\varepsilon \int_{t_0}^t \|A(x(s)) - A(\xi(s))\| \|\varphi(s, u(s))\| \Delta s}_{B_3} + \underbrace{\varepsilon \int_{t_0}^t \|A(\xi(s))\| \|\varphi(s, u(s)) - v(s)\| \Delta s}_{B_4}. \end{aligned}$$

The estimates for the first two terms in the last expression were obtained in [18] in proving an analog of the first Bogolyubov theorem for systems on time scales. These estimates take the form

$$B_1 \leq \varepsilon \lambda \int_{t_0}^t \|x(t) - \xi(t)\| \Delta s,$$

$$B_2 \leq \frac{2\lambda M L^2}{N} + 2NF_1\left(\frac{L}{\varepsilon\delta}\right),$$

where N is the number of points of the δ -partition and $F_1(T)$ is a function decreasing to zero as $T \rightarrow \infty$. The existence of this function is guaranteed by the uniform convergence to limits in (5).

In view of the Lipschitz property of the matrix $A(x)$ and the boundedness of the function $\varphi(t, u)$, the third term can be estimated as follows:

$$B_3 \leq \varepsilon \int_{t_0}^t \lambda_A M \|x(s) - \xi(s)\| \Delta s.$$

In order to estimate the fourth term, we represent the integral in the form of a sum of integrals. In view of the notation $\xi_i = \xi(t_i)$, we get

$$\begin{aligned} B_4 &= \varepsilon \int_{t_0}^t \|A(\xi(s))[\varphi(s, u(s)) - v(s)]\| \Delta s \\ &= \sum_{i \in I} \varepsilon \int_{t_i}^{t_{i+1}} \|A(\xi(s))[\varphi(s, u(s)) - v(s)] - A(\xi_i)[\varphi(s, u(s)) - v(s)] \\ &\quad + A(\xi_i)[\varphi(s, u(s)) - v(s)]\| \Delta s \\ &\leq \varepsilon \sum_{i \in I} \int_{t_i}^{t_{i+1}} \| [A(\xi(s)) - A(\xi_i)] [\varphi(s, u(s)) - v(s)] \| \Delta s \\ &\quad + \varepsilon \sum_{i \in I} \int_{t_i}^{t_{i+1}} \| A(\xi_i) [\varphi(s, u(s)) - v(s)] \| \Delta s. \end{aligned}$$

Thus, by virtue of the uniform convergence to the average value, there exists a decreasing function $F_2(T) \xrightarrow{T \rightarrow \infty} 0$ such that

$$\|\varphi(s, u(s)) - v(s)\| \leq F_2\left(\frac{L}{\varepsilon\delta}\right).$$

We now estimate the norm of the difference $\|\xi(s) - \xi_i\|$:

$$\|\xi(s) - \xi_i\| \leq \left\| \varepsilon \int_{t_i}^s [\bar{f}(\xi(\tau)) + A(\xi(\tau))v(\tau)] \Delta\tau \right\| \leq \varepsilon \int_{t_i}^s (M + M^2) \Delta\tau.$$

For B_4 , we find

$$\begin{aligned} B_4 &\leq \varepsilon \sum_{i \in I} \int_{t_i}^{t_{i+1}} \lambda \|\xi(s) - \xi_i\| \|\varphi(s, u(s)) - v(s)\| \Delta s \\ &\quad + \varepsilon \sum_{i \in I} M \int_{t_i}^{t_{i+1}} \|\varphi(s, u(s)) - v(s)\| \Delta s \\ &\leq \varepsilon \sum_{i \in I} \int_{t_i}^{t_{i+1}} \lambda \varepsilon (M + M^2) \int_{t_i}^s \Delta\tau F_2\left(\frac{L}{\delta\varepsilon}\right) \Delta s + \varepsilon M \int_{t_0}^t F_2\left(\frac{L}{\delta\varepsilon}\right) \Delta s \\ &\leq \lambda \varepsilon^2 (M + M^2) F_2\left(\frac{L}{\delta\varepsilon}\right) \sum_{i \in I} \int_{t_i}^{t_{i+1}} \int_{t_i}^s \Delta\tau \Delta s + \varepsilon M F_2\left(\frac{L}{\delta\varepsilon}\right) \frac{L}{\varepsilon} \\ &\leq \lambda \varepsilon^2 (M + M^2) F_2\left(\frac{L}{\delta\varepsilon}\right) \sum_{i \in I} \int_{t_i}^{t_{i+1}} (s - t_i) \Delta s + M F_2\left(\frac{L}{\delta\varepsilon}\right) L. \end{aligned}$$

In order to estimate the integrals in the last sum, we consider two cases. If $i \in I_\delta$, then $t_{i+1} - t_i \leq \delta$. Therefore,

$$\int_{t_i}^{t_{i+1}} (s - t_i) \Delta s \leq \delta(t_{i+1} - t_i) \leq \delta^2.$$

At the same time, if $i \in I_\sigma$, then $t_{i+1} = \sigma(t_i)$ and

$$\int_{t_i}^{t_{i+1}} (s - t_i) \Delta s = \int_{t_i}^{\sigma(t_i)} (s - t_i) \Delta s = \mu(t_i)(t_i - t_i) = 0.$$

We now use condition (v) of the theorem. Let $N_0 = \frac{L}{\varepsilon\mu_0}$. Then, for all $N > N_0$ and $\delta = \frac{L}{N\varepsilon}$, we get $\delta < \mu_0$, and it becomes possible to estimate the sum of integrals in the following way:

$$\sum_{i \in I} \int_{t_i}^s (s - t_i) \Delta s \leq \sum_{i \in I_\delta} \delta^2 = \sum_{i \in I_\delta} \frac{L^2}{N^2 \varepsilon^2} < \frac{L^2}{N \varepsilon^2}.$$

Returning to the estimation of B_4 , we arrive at the inequality

$$B_4 \leq \lambda(M + M^2) F_2 \left(\frac{L}{\delta \varepsilon} \right) \frac{L^2}{N} + M F_2 \left(\frac{L}{\delta \varepsilon} \right) L = \left(\frac{\lambda(M + M^2) L^2}{N} + ML \right) F_2 \left(\frac{L}{\delta \varepsilon} \right).$$

Finally, we get the following estimate for the norm of the difference of the trajectories of motion of systems (1) and (3):

$$\begin{aligned} \|x(t) - \xi(t)\| &\leq \varepsilon \lambda \int_{t_0}^t \|x(s) - \xi(s)\| \Delta s + \frac{2\lambda M L^2}{N} + 2N F_1 \left(\frac{L}{\delta \varepsilon} \right) \\ &\quad + \varepsilon \lambda_A M \int_{t_0}^t \|x(s) - \xi(s)\| \Delta s + \left(\frac{\lambda L^2 (M + M^2)}{N} + ML \right) F_2 \left(\frac{L}{\delta \varepsilon} \right). \end{aligned}$$

Gathering the terms in the last expression in a proper way and applying the Gronwall lemma, we obtain

$$\|x(t) - \xi(t)\| \leq \left[\frac{2\lambda M L^2}{N} + 2N F_1 \left(\frac{L}{\delta \varepsilon} \right) + \left(\frac{\lambda L^2 (M + M^2)}{N} + ML \right) F_2 \left(\frac{L}{\delta \varepsilon} \right) \right] e_{\varepsilon(\lambda + \lambda_A M)}(t, t_0).$$

In view of the monotonicity of the exponential function [15] (Theorem 2.36, item ii), we find

$$e_{\varepsilon(\lambda + \lambda_A M)}(t, t_0) < e_{\varepsilon(\lambda + \lambda_A M)} \left(\frac{L}{\varepsilon}, t_0 \right).$$

Further, since $1 + \varepsilon(\lambda + \lambda_A M)\mu(t) > 0$, in view of the definition of the exponential function via the cylindrical transformation [15, p. 59] (Definition 2.30), we obtain

$$\begin{aligned} e_{\varepsilon(\lambda + \lambda_A M)} \left(\frac{L}{\varepsilon}, t_0 \right) &= \exp \left(\int_{t_0}^{\frac{L}{\varepsilon}} \frac{\ln(1 + \varepsilon(\lambda + \lambda_A M)\mu(\tau))}{\mu(\tau)} \Delta \tau \right) \\ &< \exp \left(\int_{t_0}^{\frac{L}{\varepsilon}} \frac{\varepsilon(\lambda + \lambda_A M)\mu(\tau)}{\mu(\tau)} \Delta \tau \right) \end{aligned}$$

$$= \exp \left(\varepsilon (\lambda + \lambda_A M) \int_{t_0}^{\frac{L}{\varepsilon}} \Delta \tau \right) < e^{(\lambda + \lambda_A M)L}.$$

We estimate the exponential function as shown above and choose a sufficiently large N such that the following inequality is satisfied:

$$\frac{2\lambda ML^2}{N} e^{(\lambda + \lambda_A M)L} < \frac{\eta}{3}.$$

Further, we fix the indicated value of N and choose ε_0 such that the following inequalities are satisfied for all $\varepsilon \in (0, \varepsilon_0)$:

$$2NF_1 \left(\frac{L}{\delta\varepsilon} \right) e^{(\lambda + \lambda_A M)L} < \frac{\eta}{3},$$

$$F_2 \left(\frac{L}{\delta\varepsilon} \right) \left(\frac{\lambda L^2(M + M^2)}{N} + ML \right) e^{(\lambda + \lambda_A M)L} < \frac{\eta}{3}.$$

Finally, we arrive at the estimate

$$\|x(t) - \xi(t)\| < \eta.$$

Thus, the first assertion of the theorem is proved.

As follows from the description of the algorithm of correspondence of controls, the proof of the second assertion of the theorem is similar.

Theorem 1 is proved.

Remark 5. Note that, from the practical viewpoint, the first part of the algorithm is of especial interest. It enables one to associate any control of the averaged problem (system (3) is autonomous and, hence, it is regarded as a simpler system) with an admissible control in the original problem (1), (2). Theorem 1 allows us to state that the corresponding trajectories of motion of the systems are close to each other. The second part of the algorithm has only theoretical significance.

By using the method of averaging for the equations of controlled motion, one can easily substantiate the numerical-asymptotic method for the solution of the problem of optimal control with terminal quality criterion (1), (2). For this purpose, we first prove several lemmas to establish the relationships between the problems of optimal control with the same quality criterion.

Thus, we consider the following two problems of optimal control:

$$\begin{aligned} x^\Delta(t) &= f(t, x, u), \\ x(t_0) &= x_0, \end{aligned} \quad J[u] = \Phi(x(T)) \rightarrow \min_{u \in U}, \quad (7)$$

$$\begin{aligned} y^\Delta(t) &= g(t, y, v), \\ y(t_0) &= x_0, \end{aligned} \quad \bar{J}[v] = \Phi(y(T)) \rightarrow \min_{v \in V}, \quad (8)$$

where $x \in \mathbb{R}^n$, $y \in \mathbb{R}^n$, $f(t, x, u)$ and $g(t, y, u)$ are n -dimensional vector functions, $\Phi(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Lipschitz function with Lipschitz constant λ_Φ , $u(t) \in U$ an r -dimensional control vector for problem (7), $U \in \text{comp}(R^r)$, $v(t) \in V$ is an s -dimensional control vector (8), and $V \in \text{comp}(R^s)$.

Assume that problems (7) and (8) are solvable (possibly not uniquely). We denote the optimal controls for problems (7) and (8) by u^* and v^* , respectively.

Lemma 2. *Assume that there exists an admissible control $v^0 \in V$ such that the trajectory corresponding to this control $y(t)$ is close to the optimal trajectory $x^*(t)$ of problem (7), i.e., for small $\eta_1 > 0$ and any $t \in [t_0, T]$, the inequality $\|x^*(t) - y(t)\| < \eta_1$ is true. Also let $J[u^*] \leq \bar{J}[v^*]$. Then*

$$|J[u^*] - \bar{J}[v^*]| < \lambda_\Phi \eta_1.$$

Proof. By the Lipschitz property of $\Phi(\cdot)$, in view of the special choice of v^0 , we get

$$|J[u^*] - \bar{J}[v^0]| = |\Phi(x^*(T)) - \Phi(y(T))| \leq \lambda_\Phi |x^*(T) - y(T)| \leq \lambda_\Phi \eta_1.$$

In other words,

$$J[u^*] \geq \bar{J}[v^0] - \lambda_\Phi \eta_1. \quad (9)$$

Since v^* is the optimal control of problem (8), we find

$$\bar{J}[v^*] \leq \bar{J}[v^0]. \quad (10)$$

In view of (9) and (10), we arrive at the following chain of inequalities:

$$\bar{J}[v^0] \geq \bar{J}[v^*] \geq J[u^*] \geq \bar{J}[v^0] - \lambda_\Phi \eta_1.$$

Subtracting $\bar{J}[v^*]$ from this inequality, we obtain

$$J[u^*] - \bar{J}[v^*] \geq \bar{J}[v^0] - \bar{J}[v^*] - \lambda_\Phi \eta_1.$$

Hence, by virtue of (10), the following inequality is also true:

$$J[u^*] - \bar{J}[v^*] \geq -\lambda_\Phi \eta_1.$$

On the other hand, by the condition of the lemma, we get

$$J[u^*] - \bar{J}[v^*] \leq 0 < \lambda_\Phi \eta_1.$$

Thus,

$$-\lambda_\Phi \eta_1 \leq J[u^*] - \bar{J}[v^*] < \lambda_\Phi \eta_1.$$

Lemma 2 is proved.

Lemma 3. Assume that there exists a control u^0 of problem (7) such that the trajectory $x(t)$ corresponding to this control is close to the optimal trajectory $y^*(t)$ of problem (8), i.e., the inequality $\|x(t) - y^*(t)\| < \eta_2$ holds for small $\eta_2 > 0$ and any $t \in [t_0, T]$. Also let $J[u^*] > \bar{J}[v^*]$. Then

$$|J[u^*] - \bar{J}[v^*]| < \lambda_\Phi \eta_2.$$

Proof. As in the proof of Lemma 2, we obtain the chain of inequalities

$$J[u^0] \geq J[u^*] > \bar{J}[v^*] \geq J[u^0] - \lambda_\Phi \eta_2.$$

Subtracting $J[u^*]$ from this inequality, we get

$$\bar{J}[v^*] - J[u^*] \geq J[u^0] - J[u^*] - \lambda_\Phi \eta_2.$$

Thus, the following inequality is also true:

$$\bar{J}[v^*] - J[u^*] \geq -\lambda_\Phi \eta_2.$$

In other words,

$$J[u^*] - \bar{J}[v^*] \leq \lambda_\Phi \eta_2.$$

On the other hand, by the condition of the lemma, we find

$$J[u^*] - \bar{J}[v^*] > 0 > -\lambda_\Phi \eta_2$$

and, therefore,

$$-\lambda_\Phi \eta < J[u^*] - \bar{J}[v^*] \leq \lambda_\Phi \eta_2.$$

Lemma 3 is proved.

Lemma 4. Assume that the admissible controls u^0 and v^0 from Lemmas 2 and 3 simultaneously exist. Then

$$|J[u^*] - \bar{J}[v^*]| < \lambda_\Phi \eta_3,$$

$$J[u^0] - J[u^*] \leq C \eta_3,$$

$$\bar{J}[v^0] - \bar{J}[v^*] \leq C \eta_3,$$

where $\eta_3 = \max\{\eta_1, \eta_2\}$.

Proof. One of the indicated two quantities ($J[u^*]$ or $\bar{J}[v^*]$) is necessarily larger (or at least equal) than the other quantity. Hence, either Lemma 2 or Lemma 3 is true and, in any case, the following inequality holds:

$$|J[u^*] - \bar{J}[v^*]| < \lambda_\Phi \eta_3.$$

Therefore,

$$\begin{aligned} J[u^0] - J[u^*] &= J[u^0] - \bar{J}[v^*] + \bar{J}[v^*] - J[u^*] \\ &\leq |J[u^0] - \bar{J}[v^*]| + |\bar{J}[v^*] - J[u^*]| \leq \lambda_\Phi \eta_2 + \lambda_\Phi \eta_3 < C \eta_3, \end{aligned}$$

where $C = 2\lambda_\Phi$.

Lemma 4 is proved.

Theorem 2. *Assume that the conditions of Theorem 1 are satisfied in the domain*

$$Q = \{t \in \mathbb{T}, x \in D \subset \mathbb{R}^n, u \in U \subset \mathbb{R}^r\}$$

and, in addition,

- (i) $\Phi(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Lipschitz function with Lipschitz constant λ_Φ ;
- (ii) both the optimal control $u^*(t) \in U$ of problem (1), (2) and the optimal control $v^*(t) \in V$ of problem (3), (4) exist.

Then, for any $0 < \eta \leq \rho$ and $L > 0$, there exists $\varepsilon_0(\eta, L) > 0$ such that the following inequalities hold for $0 < \varepsilon < \varepsilon_0$ and $t_0 \leq t \leq L\varepsilon^{-1}$:

$$|J[u^*] - \bar{J}[v^*]| < \eta,$$

$$J[u_{v^*}] - J[u^*] < \eta,$$

where u_{v^*} is a control of system (1) constructed according to the algorithm and associated with the optimal control v^* of problem (3), (4).

Proof. In order to prove the theorem, it suffices to note that the algorithm of the method of averaging enables us to associate the control v^* with an admissible control u_{v^*} of problem (1), (2) such that the corresponding trajectories are close:

$$\|x_{v^*}(t) - \xi^*(t)\| < \eta.$$

By ε_1 we denote the value of small parameter ε , for which the last inequality is satisfied according to Theorem 1.

On the other hand, the same algorithm allows us to construct a control u_{u^*} for the averaged system corresponding to the optimal control of the original system. By Theorem 1, there exists ε_2 such that the corresponding trajectories are also close:

$$\|x^*(t) - \xi_{u^*}(t)\| < \eta.$$

Applying Lemma 4, we immediately obtain both inequalities of the theorem.

Theorem 2 is proved.

Theorem 2 completes the substantiation of the numerical-asymptotic method for the solution of the problem of optimal control (1), (2).

REFERENCES

1. N. M. Krylov and N. N. Bogolyubov, *Introduction to Nonlinear Mechanics*, Princeton Univ. Press, Princeton (1947).
2. N. N. Bogolyubov, Yu. A. Mitropol'skii, *Asymptotic Methods in the Theory of Nonlinear Oscillations*, Gordon & Breach, Delhi (1961).
3. N. N. Moiseev, *Asymptotic Methods in Nonlinear Mechanics* [in Russian], Nauka, Moscow (1981).
4. I. A. Boitsova, "Numerical-asymptotic method for the solution of discrete problems of optimal control with fast and slow variables," *Vestn. Belor. Gos. Univ. Ser. I*, No. 1, 105–110 (2011).
5. T. A. Komleva, L. I. Plotnikova, A. V. Plotnikov, and N. V. Skripnik, "Averaging in fuzzy controlled systems," *Nelin. Kolyv.*, **14**, No. 3, 325–332 (2011); **English translation:** *Nonlin. Oscillat.*, **14**, No. 3, 342–349 (2012).
6. A. V. Plotnikov, "Asymptotic investigation of the equations of controlled motion with multivalued trajectories," *Ukr. Mat. Zh.*, **42**, No. 10, 1409–1412 (1990).
7. V. A. Plotnikov, A. V. Plotnikov, and A. N. Vityuk, *Differential Equations with Multivalued Right-Hand Side. Asymptotic Methods* [in Russian], Astroprint, Odessa (1999).
8. V. A. Plotnikov, *Method of Averaging in the Problems of Control* [in Russian], Lybid', Kiev, Odessa (1992).
9. V. A. Plotnikov, L. I. Plotnikova, and A. T. Yarovoi, "Averaging method for discrete systems and its application to control problems," *Nelin. Kolyv.*, **7**, No. 2, 241–254 (2004); **English translation:** *Nonlin. Oscillat.*, **7**, No. 2, 240–253 (2004).
10. V. A. Plotnikov and L. I. Plotnikova, "Averaging of the equations of controlled motion in metric spaces," *Kibernet. Syst. Anal.*, No. 4, 175–180 (1997).
11. V. A. Plotnikov and O. D. Kichmarenko, "Averaging of controlled equations with Hukuhara derivative," *Nelin. Kolyv.*, **9**, No. 3, 376–385 (2006); **English translation:** *Nonlin. Oscillat.*, **9**, No. 3, 365–374 (2006).
12. V. A. Plotnikov and O. D. Kichmarenko, "Averaging of the equations with Hukuhara derivative, multivalued control, and delay," *Visn. Odes. Nat. Univ., Mat. Mekh.*, **12**, Issue 7, 130–139 (2007).
13. A. V. Plotnikov, "The averaging of control linear fuzzy differential equations," *J. Adv. Res. Appl. Math.*, **3**, No. 3, 1–20 (2011).
14. S. Hilger, *Ein Maßkettenkalkül mit Anwendung auf Zentrumsmannigfaltigkeiten*, Ph.D. Thesis, Univ. Würzburg, Würzburg (1988).
15. M. Bohner and A. C. Peterson, *Dynamic Equations on Time Scales: an Introduction with Applications*, Birkhäuser, Basel (2001).
16. M. Bohner and A. C. Peterson, *Advances in Dynamic Equations on Time Scales*, Springer, Berlin (2003).
17. A. Slavík, "Averaging dynamic equations on time scales," *J. Math. Anal. and Appl.*, **388**, No. 2, 996–1012 (2012).
18. O. P. Ogulenko and O. D. Kichmarenko, "A scheme of complete averaging on time scales," *Visn. Odes. Nat. Univ. Mat. Mekh.*, **17**, Issue 4 (16), 67–77 (2012).